

HDOC Day-15 Assignment

AIM : write a menu – driven C code using switch-case to input a positive integer and perform operations based on user choice .

1. Check whether all digits of the number are even.
2. Check whether all digits of the number are prime (prime digits are 2, 3, 5, 7).

Algorithm ;

- Start.
- Display menu options to the user:
- Choice 1: Check if all digits are even.
- Choice 2: Check if all digits are prime.
- Read the user's choice.
- Read a positive integer num.
- Perform actions based on the choice using a switch-case statement:
- **Case 1 (Check Even Digits):**
 - Copy num to a temporary variable temp.
 - Initialize a flag allEven = 1.
 - While temp > 0:
 - Find the last digit using digit = temp % 10.
 - Check if digit % 2 != 0. If yes, set allEven = 0 and break.
 - Reduce temp using temp = temp / 10.
 - If allEven == 1, print "All digits are even".
 - Else, print "All digits not even".
- **Case 2 (Check Prime Digits):**
 - Copy num to a temporary variable temp.
 - Initialize a flag allPrime = 1.
 - While temp > 0:
 - Find the last digit using digit = temp % 10.
 - Check if digit is not equal to 2, 3, 5, or 7. If true, set allPrime = 0 and break.
 - Reduce temp using temp = temp / 10.
 - If allPrime == 1, print "All digits are prime".
 - Else, print "All digits are not prime".

- **Default Case:**
 - Print "Invalid choice".

TEST CASE TABLES :

Test case no	choice	Input number	Expected op	Actual op	status
1.	1	4682	All digits are even	All digits are even	pass
2.	1	4672	All digits not even	All digits not even	pass
3.	2	2357	All digits are prime	All digits are prime	pass
4.	2	3589	All digits are not prime	All digits are not prime	pass

C Program Code

```
#include <stdio.h>
```

```
int main() {
```

```
    int choice;
```

```
    int num, temp, digit;
```

```
    int allEven = 1;
```

```
    int allPrime = 1;
```

```
    printf("1. Check if all digits are even\n");
```

```
    printf("2. Check if all digits are prime\n");
```

```
    printf("Enter your choice (1 or 2): ");
```

```
scanf("%d", &choice);
```

```
printf("Enter a positive number: ");
```

```
scanf("%d", &num);
```

```
switch(choice) {
```

```
    case 1:
```

```
        temp = num;
```

```
        while(temp > 0) {
```

```
            digit = temp % 10;
```

```
            if(digit % 2 != 0) {
```

```
                allEven = 0;
```

```
                break;
```

```
            }
```

```
            temp = temp / 10;
```

```
        }
```

```
        if(allEven == 1) {
```

```
            printf("All digits are even\n");
```

```
        } else {
```

```
            printf("All digits not even\n");
```

```
        }
```

```
        break;
```

case 2:

```
temp = num;
while(temp > 0) {
    digit = temp % 10;
    if(digit != 2 && digit != 3 && digit != 5 && digit != 7) {
        allPrime = 0;
        break;
    }
    temp = temp / 10;
}
```

```
if(allPrime == 1) {
    printf("All digits are prime\n");
} else {
    printf("All digits are not prime\n");
}
break;
```

default:

```
printf("Invalid choice!\n");
break;
}
```

```
return 0;
```

```
}
```

```
#include <stdio.h>

int main() {
    int choice;
    int num, temp, digit;
    int allEven = 1;
    int allPrime = 1;

    printf("1. Check if all digits are even\n");
    printf("2. Check if all digits are prime\n");
    printf("Enter your choice (1 or 2): ");
    scanf("%d", &choice);

    printf("Enter a positive number: ");
    scanf("%d", &num);

    switch(choice) {
        case 1:
            temp = num;
            while(temp > 0) {
                digit = temp % 10;
                if(digit % 2 != 0) {
                    allEven = 0;
                    break;
                }
                temp = temp / 10;
            }

            if(allEven == 1) {
                printf("All digits are even\n");
            } else {
                printf("All digits not even\n");
            }
            break;
```

```
    case 2:
        temp = num;
        while(temp > 0) {
            digit = temp % 10;
            if(digit != 2 && digit != 3 && digit != 5 && digit != 7) {
                allPrime = 0;
                break;
            }
            temp = temp / 10;
        }

        if(allPrime == 1) {
            printf("All digits are prime\n");
        } else {
            printf("All digits are not prime\n");
        }
        break;

    default:
        printf("Invalid choice!\n");
        break;
}

return 0;
}
```

Program Output Execution

```
[manalithakar@Manalis-MacBook-Air ~ % gcc 15q1.c -o 15q1
[manalithakar@Manalis-MacBook-Air ~ % ./15q1
1. Check if all digits are even
2. Check if all digits are prime
Enter your choice (1 or 2): 1
Enter a positive number: 4682
All digits are even
[manalithakar@Manalis-MacBook-Air ~ % ./15q1
1. Check if all digits are even
2. Check if all digits are prime
Enter your choice (1 or 2): 1
Enter a positive number: 4672
All digits not even
[manalithakar@Manalis-MacBook-Air ~ % ./15q1
1. Check if all digits are even
2. Check if all digits are prime
Enter your choice (1 or 2): 2
Enter a positive number: 2357
All digits are prime
[manalithakar@Manalis-MacBook-Air ~ % ./15q1
1. Check if all digits are even
2. Check if all digits are prime
Enter your choice (1 or 2): 2
Enter a positive number: 3589
All digits are not prime
manalithakar@Manalis-MacBook-Air ~ % █
```